

Project Overview: ITU (International Telecom Utility) Platform

This document provides a comprehensive overview of the **ITU (International Telecom Utility)** platform. It is structured to serve as a primary reference source for Retrieval-Augmented Generation (RAG) systems, detail user-facing features, explain transaction flows, and outline the technical architecture.

1. Project Mission & Objectives

The core aim of the ITU platform is:

"Connecting families across borders through instant, secure, and reliable mobile recharge."

ITU serves as a digital bridge between diaspora communities (living in countries like Germany, the USA, the UK, Canada, Australia, etc.) and their loved ones in over **180+ countries** worldwide. The platform enables users to send airtime, mobile data, SMS bundles, and unified telecommunication packages directly to any prepaid mobile phone number.

Core Value Propositions

- **Instant Processing:** Over 99% of top-up transfers are completed within seconds.
- **Broadest Global Reach:** Support for **700+ mobile network**

operators across Africa, Asia, Europe, Latin America, and the Middle East.

- **Cost Efficiency:** A Least Cost Routing (LCR) system dynamically scans aggregators to offer the lowest possible prices and service charges.
 - **Enterprise Security:** Secure payments processed through Razorpay with 128-bit SSL transaction encryption and strict Row Level Security (RLS) database policies.
 - **Customer Retention:** An integrated wallet system, custom recurring top-ups, and a loyalty rewards program.
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2. Supported Regions & Usage Scenarios

Where the User Can Use ITU

ITU is designed as a cross-border mobile web application.

- **Sender Countries (Origins):** Primarily accessed by users residing in Germany, Western Europe, North America (USA, Canada), and Australia.
- **Recipient Countries (Destinations):** Over 180+ countries, with major active channels in India (IND), Nigeria (NGA), the Philippines (PHL), Brazil (BRA), Mexico (MEX), and various countries across Asia, Europe, South America, and Africa.
- **Access Modes:** Accessible via any modern web browser on desktop, tablet, or smartphone.

Key Use Cases

1. **Family Support:** Sending monthly mobile data and airtime allowances to aging parents or siblings back home.
2. **Gift Exchanges:** Sending airtime or digital top-up vouchers as gifts during holidays or festivals.
3. **Remote Payment of Utilities:** Adding credits to international SIM cards for remote workers or travelers.

4. **Emergency Credits:** Instantly sending top-ups when a family member runs out of mobile data in an emergency.
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3. User-Facing Features

The ITU platform delivers a comprehensive suite of utility and transactional services:

3.1. International Mobile Recharge (Top-Ups)

- **Airtime Top-ups:** Instantly send monetary balance directly to a prepaid mobile number.
- **Mobile Data Packages:** Choose specific GB packages tailored to the network carrier (e.g., 5GB daily, weekly social media packs).
- **SMS & Voice Bundles:** Operator-specific packages covering designated local and international calling minutes or text counts.
- **Combo Plans:** Combined data, voice, and SMS offerings.
- **Operator Identification:** Automatic detection of the recipient's country code and telecom carrier based on prefix lookup, with the option for manual correction.

3.2. Digital Wallet

- **Balance Storage:** Registered users have a personal multi-currency wallet (defaulting to USD or EUR).
- **Quick Checkout:** Users can recharge their wallet balance and use it to instantly purchase top-ups, bypassing payment gateway input screens.
- **Partial Wallet Payments:** Users can combine their existing wallet balance with card payments to cover orders.

3.3. Recurring Recharges

- **Automated Schedules:** Users can schedule recharges to trigger

automatically at set intervals.

- **Supported Frequencies:** Weekly, Bi-weekly, Monthly, or a custom interval in days.
- **Auto-Billing:** Automates payments through saved payment methods or wallet deductions.

3.4. Loyalty Rewards Program

- **Points Accumulation:** Users earn reward points based on the value of their transactions (e.g., earning points per Euro/Dollar spent).
- **Redemption:** Points can be redeemed at checkout for discounts or converted directly into wallet balance.

3.5. Digital Vouchers

- **Code Purchases:** Buy digital codes/PINs for entertainment, internet, or operator utilities.
- **Redemption Flow:** Vouchers can be purchased and sent to recipient emails or redeemed within the checkout flow to apply discounts.

3.6. Interactive Help & Support

- **Ticket Management:** Users can submit support tickets for failed payments, wrong numbers, or refund requests.
 - **Interactive Chatbot:** An automated assistant (`support-bot`) offers immediate troubleshooting, matches ticket subjects to FAQs, and provides suggestions.
 - **Receipt Downloads:** Generation of formal PDF invoices and transaction receipts available on the success page and user dashboard history.
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4. How to Recharge: User Guide Step-by-

Step

Recharging a mobile phone on the ITU platform follows a highly optimized, 4-step wizard-style flow:

```
graph TD
    A[Step 1: Recipient Phone Number] -->|Auto-detects Co| B
    B -->|Selects Airtime, Data, or Voice Pack| C[Step 3:
    C -->|Choose Wallet or Razorpay Checkout| D[Step 4: S
    D -->|Download PDF Receipt & Earn Rewards| E[Done]
```

Step 1: Recipient Information Entry

1. The user navigates to the **Recharge** page (`/recharge`).
2. The user enters the recipient's mobile number in international format (e.g., +919876543210 for India).
3. The platform executes a prefix analysis to identify the destination **Country** and **Mobile Operator** (e.g., Airtel, MTN, Globe, Jio).
4. The user verifies the detected carrier. If incorrect (due to mobile number portability), the user can click **Change Carrier** to manually choose from the list of operators in that country.

Step 2: Selecting the Recharge Plan

1. Once the carrier is selected, the platform fetches a normalized list of plans (categorized into *Data*, *Voice*, *Combo*, and *Airtime*).
2. The user reviews the plan benefits (e.g., "1.5 GB/Day, Unlimited Calls, 28 Days Validity"), pricing in their local currency (EUR/USD), and the exact credit the recipient will receive in local currency (e.g., INR, NGN, PHP).
3. The user clicks **Select Plan** to proceed.

Step 3: Review & Checkout

1. The system displays a transaction breakdown:

- **Recharge Amount:** The base price of the plan.
 - **Service Fee:** A minimal system maintenance charge (e.g., \$0.50).
 - **Wallet Discount:** Deductions if the user chooses to redeem reward points or wallet balance.
 - **Total Payable:** The final payment amount.
2. The user selects their payment option:
 - **Razorpay Checkout:** The user pays via credit/debit card, UPI, or net banking.
 - **Wallet Payment:** Pays directly from their prepaid ITU wallet.
 3. The user enters an optional **Idempotency Key** (auto-generated in background) to guarantee that they are not double-charged if their connection drops.
 4. The user completes the payment transaction.

Step 4: Success Confirmation

1. Upon validation, the user is redirected to the **Success** page.
 2. The success page displays:
 - Operator reference ID (proving the aggregator successfully credited the recipient).
 - Order status ("completed").
 - Reward points earned during the purchase.
 - A button to **Download PDF Receipt**.
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7. Frequently Asked Questions (FAQ)

Q1: How long does a mobile top-up take to credit the recipient?

Most mobile recharges complete instantly (under 10 seconds). In rare scenarios (e.g., operator network congestion), it may take up to a few minutes. If a delay exceeds 30 minutes, users should contact support.

Q2: I entered the wrong phone number. Can I get a refund?

Mobile recharges are processed instantly through national carrier networks. Once processed, transactions cannot be reversed or refunded. Users must double-check the country prefix and number before submitting payment.

Q3: How do digital vouchers work on the platform?

Digital vouchers generate unique redemption PINs. During checkout, users can enter a voucher code to apply discount credits. Vouchers can also be purchased as gifts and emailed to recipients.

Q4: How do I access and download invoice receipts?

Upon successful recharge, a receipt download button is displayed on the confirmation page. Additionally, registered users can view their history in the **Account Dashboard** under the **Transactions** tab to download PDF receipts for any past order.

Q5: What payment methods does ITU support?

ITU supports credit cards, debit cards, UPI, net banking, and direct payments from the user's in-app digital wallet. Credit/debit card and UPI payments are secured via Razorpay.

Q6: How does the loyalty rewards program work?

Every time you complete a recharge, you earn reward points calculated from the recharge amount. These points are stored in your account profile and can be redeemed during checkout for discounts on future recharges.